Yuankun Huang

Software Engineer, Game Systems & Simulation

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Unity/C# engineer: five years on mobile games (two shipped titles, one from early development to 10M+ downloads) and a year as sole engineer rebuilding a clinically validated rehabilitation game. Working toward real-time simulation in C++20.

CORE SKILLS

Languages	C# (5+ years in production), C++20
Engines & Runtime	Unity (mobile, live service), Addressables, ILRuntime hot-update, Unity Profiler, .NET 8
Systems	State sync (Protobuf, WebSocket, SignalR, MessagePack), deterministic replay, multi-agent simulation
Build & Quality	CMake, GitHub Actions CI, custom allocation guards

PROFESSIONAL EXPERIENCE

Unity Developer May 2025 – Present
IGG Canada, live-service and new-title client development *Vancouver, BC*

- Own the live-event client framework of an unannounced strategy title (C#/Unity), from scheduling and entry resolution to the shared hub UI; six other engineers ship 30+ event types on it.
- Built 13 of those event types end to end (guild duel, boss hunts, novice and battle passes), from data parser to UI.
- Built a build-size strip pass with a visual regression check for the hero showcase scenes (Timeline/Animator): removes never-shown prefab subtrees as reversible overrides, compares full and stripped frame by frame, rolls back on any visible loss.
- Wrote automated Play Mode regression suites for 12 live-event types (78 cases) on the team's Editor test framework, plus shared test utilities (server-time forwarding) and a scheduled nightly runner.
- Wrote the module's layering standard and a prefab validator every generator saves through; mentor new client engineers.
- Shipped Mythic Heroes' 2025 live-event wave (pet progression, event pass, 25 UI windows) as monthly ILRuntime hot-updates.

Career Break: Bereavement & Family Care *Feb 2025 – May 2025*

Software Engineer (Contract) Jan 2024 – Jan 2025
The HIVE, UBC Faculty of Medicine, gamified stroke rehabilitation *Vancouver, BC*

- Sole engineer on the Unity rebuild of TrAIT, a stroke-rehabilitation game with Ultraleap hand tracking: scene and manager architecture, multi-stage calibration, and streak-based adaptive levels, with a product owner and a UX designer.
- Built the research telemetry (per-tick CSV logs, emailed to the lab) and a ScriptableObject config pipeline with editor code generation; succeeds the Kinect version validated in *Neurorehabilitation and Neural Repair* (13% arm-function gain).

Unity Developer Sep 2020 – Sep 2024
IGG Canada, Mythic Heroes (live-service mobile RPG, 10M+ downloads) *Vancouver, BC*

- Core client engineer from early development to global launch; owned the Chat client module and co-owned Guilds, including client networking and multiplayer state sync over Google Protobuf; shipped monthly content via ILRuntime hot-updates.
- Built the client network module for Dawn of Survival, a second shipped title: WebSocket transport, Protobuf messages, gzip HTTP session layer, and login adapters, adapted from the flagship's proven stack.
- Co-developed the shared UI framework and authored 150+ of its windows; profiled and optimized for low-end Android with the Unity Profiler and Frame Debugger (off-screen list culling, leaderboard caching, sprite-atlas migration to Addressables).

SELECTED PROJECTS

Rhythm Fruit Shop [GitHub] *C++20 / SFML / CMake / GitHub Actions*

- C++20/CMake rhythm core; GitHub Actions CI enforces that deterministic replay reproduces the recorded result exactly and a perfect run matches a reference scoring implementation; zero-allocation hot path asserted by a custom allocation guard.

Kodama [GitHub] *.NET 8 / SignalR / MessagePack / Unity*

- Server-authoritative multi-agent simulation (10K agents) in .NET 8 (SoA component storage, zero-allocation behaviour pass); state synced via SignalR and MessagePack to a GPU-instanced Unity client.

EDUCATION & CERTIFICATION

M.S. Computer Science (in progress, part-time), Georgia Institute of Technology 2026 – Present
Master of Digital Media, Centre for Digital Media, Vancouver 2021
B.Eng. Telecommunication Engineering, Queen Mary University of London (joint programme with BUPT) 2019
Unity Certified Expert: Programmer